SYNC A/V — TALK & WORKSHOP

How we perceive sound and image as one, and how to use it in live performance

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1. OVERVIEW

Sync A/V is a lecture and optional hands-on workshop by Manuel Palenque on multisensory perception and real-time audiovisual performance. It examines why the human brain fuses sound and image into a single event, and how artists, musicians, and technologists can compose with those perceptual mechanisms on stage.

FORMAT	DURATION	AUDIENCE
Talk (`Sync A/V`)	90 minutes (with video examples & Q&A)	Musicians, creative coders, visual artists, designers, media arts students
Workshop (`From Set to Stage`)	3 to 6 hours (1 or 2 sessions)	Beginner to intermediate practitioners working with real-time video, sound, and light
Live A/V Performance	40–60 minutes (*Colapso* / Live A/V)	Optional evening performance paired with the talk or workshop

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2. PART I — THE TALK: SEVEN PRINCIPLES OF MULTISENSORY INTEGRATION

Mechanisms of multisensory integration act on every audience automatically, setting the ground rules for any live audiovisual piece. Understanding why sound and image fuse turns every synchronization decision in a live set into a compositional one.

1. Synchresis — The spontaneous perceptual weld between a simultaneous auditory event and visual event, independent of physical realism.
2. The Ventriloquism Effect — Spatial capture across sensory modalities, where visual localization pulls perceived sound sources across stage architecture.
3. The McGurk Effect — Cross-modal perceptual transformation when conflicting visual and auditory cues produce a third perceived reality.
4. The Temporal Binding Window — The millisecond threshold within which asynchronous audio, video, and lighting stimuli are still perceived as unified—and how latency breaks or stretches it.
5. Common Fate — Grouping visual geometry, particle systems, and sonic trajectories that share common motion, acceleration, or decay.
6. Rhythmic Entrainment — Neural and physical synchronization to periodic audio-visual-lighting pulses in immersive environments.

7. Prediction and Reward — Composing tension, expectation, and collapse by establishing and subverting cross-modal rules in real time.

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3. PART II — THE WORKSHOP: FROM SET TO STAGE

For institutions, festivals, and media labs seeking a practical studio session alongside the talk, the workshop explores the real-time synchronization techniques used in live shows across three connected layers:

- 01 / Video: Real-time generative visuals, particle systems (vvvv / TouchDesigner), and pre-visualization in a 3D digital twin of the stage before physical load-in.
- 02 / Light: Stage lighting and DMX512 / Art-Net devices treated as an extension of the visual and sonic instrument rather than separate accompaniment.
- 03 / Sound: Audio analysis, MIDI/OSC routing, and low-latency communication between sound synthesis and visual rendering.

Participants can bring their own laptops, controllers, or instruments to test ideas directly on the live demonstration set.

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4. TECHNICAL REQUIREMENTS

- Talk (90 min): Darkened auditorium or screening room, 1080p/4K projection or LED display, stereo PA system with clean low-frequency reproduction (HDMI or balanced XLR audio from presenter laptop), and microphone for larger rooms.
- Workshop (3–6 hrs): Room with controlled darkness, projector or LED screen, stereo sound system, table for the technical set, and DMX lighting fixtures when available.

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5. ABOUT MANUEL PALENQUE

Manuel Palenque is a creative technologist and visual artist based in Buenos Aires. For over 15 years he has built real-time audiovisual performances, interactive installations, and live shows where music and image are generated together and respond to each other.

- Performed: MUTEK Buenos Aires · Tecnópolis · Beyond Code #2, Tokyo · School of Creative Media, Hong Kong · ArtHaus
- Taught: NODE Festival, Berlin · TECH LAB PAAK, Tokyo · UNTREF
- Runs: Dataraiz, a media lab in Buenos Aires, and Black Cycle, an art and technology festival

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6. CONTACT

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